CENG 252 Practice Midterm 2: Solutions

Lectures 6–10 • 100 points • 80 minutes

Problem	Final answers	Similar to
1	(a) $-2k_5C^2_{CH_3\bullet}$; (b) H_2 dissociative, A molecular, dual-site RLS, B weakly adsorbed; (c) X competitive, Y uncompetitive, only X; (d) (i) dual-site, (ii) adsorption, (iii) single-site; (e) 2nd order, $k_d = 0.025$ h^{-1}, sintering, $t = 120$ h; (f) $-r'_A = a(t)(-r'_A)_{fresh}$; (g) PBR, STTR, moving bed; (h) sintering (irreversible), S poisoning (irreversible), coking (reversible)	L6–L10
2	$-r_{O_3} = \dfrac{2k_1k_3C^2_{O_3}}{k_2C_{O_2} + k_3C_{O_3}}$; $k_{obs} = 2k_1k_3/k_2$; initially $2k_1C_{O_3}$; $E_{app} = 118$ kJ/mol, factor 4.60	L6 Ex. 1, L6 NO, HW3-2
3	uncompetitive rate law; $V_{max} = 2.0$ mmol/L/min, $K_M = 5.0$ mmol/L; uncompetitive, $K_I = 2.0$ mmol/L; $t = 14.8$ min (no I), 32.8 min (with I)	L7, HW3-3
4	(a) $\dfrac{kP_A}{(1 + K_AP_A + K_BP_B + K_WP_W)^2}$; (b) $\dfrac{k_AC_tP_A}{1 + K_BP_B + K_WP_W}$; $k = 0.50$ mol/(kg·s·atm), $K_A = 0.50$ atm^{-1}; $W = 26.7$ kg; $P_0 = 6$ atm, $W_{min} = 20.0$ kg; $a = 0.5$, $F_{A0} = 5.0$ mol/s, $T = 518.7$ K	L8–L10, HW3-4, HW3-5

Problem 1: Short answers and concepts (25 pts)

(a) Stoichiometric factor in a PSSH balance

By relative rates for $2\,CH_3^\bullet \longrightarrow C_2H_6$: $\dfrac{r_{5,CH_3\bullet}}{-2} = \dfrac{r_{5,C_2H_6}}{1}$, so

Answer

$r_{5,CH_3\bullet} = -2k_5C^2_{CH_3\bullet}$. Two radicals are destroyed for every C_2H_6 formed. This is exactly the same factor as in $2k_1C_{Br_2}$ for H_2/Br_2 initiation (Lec. 7).

(b) Reading an L–H rate law

Answer

Any three of:

1. $\sqrt{K_HP_{H_2}}$ in the denominator: **H_2 adsorbs dissociatively**, $H_2 + 2\,S \rightleftharpoons 2\,H\cdot S$.
2. K_AP_A in the denominator: **A adsorbs molecularly** on the same sites as H.
3. **Squared denominator**: the RLS needs two adjacent sites, i.e. a **dual-site surface reaction** $A\cdot S + H\cdot S \longrightarrow \ldots$, which is why the numerator contains $P_AP^{1/2}_{H_2}$ (one adsorbed A and one adsorbed H atom).
4. No P_B term in the denominator: B is **weakly adsorbed** (or desorbs instantly). No reverse term: the reaction is irreversible.

(c) Inhibitor types

Answer

X: **competitive** (same intercept $1/V_{\max}$, slope $\times(1 + C_I/K_I)$). Y: **uncompetitive** (parallel line, intercept $\times(1 + C_I/K_I)$). Only with **X** can $V_{\max}$ still be reached. As $C_S \to \infty$ ($1/C_S \to 0$), the competitive line goes to the same intercept: substrate out-competes the inhibitor for the free enzyme. With an uncompetitive inhibitor, more S just makes more E·S for I to bind, so the apparent maximum is $V_{\max}/(1 + C_I/K_I)$.

(d) Initial-rate curves

Answer

- (i) maximum: **dual-site surface reaction**, $-r'_{A0} = kP_{A0}/(1 + K_A P_{A0})^2$. At high P_{A0}, A covers the surface and there are no vacant neighbouring sites left. The maximum is at $P_{A0} = 1/K_A$.
- (ii) straight line through the origin: **adsorption of A limiting**, $-r'_{A0} = k_A C_t P_{A0}$. There is no $K_A P_A$ term, so it never saturates.
- (iii) levels off: **single-site surface reaction**, $-r'_{A0} = kP_{A0}/(1 + K_A P_{A0}) \to k/K_A$ when the surface is full.

(e) Decay order

Test both forms. The rate constant must be the same at both times:

	$t = 10$ h, $a = 0.80$	$t = 40$ h, $a = 0.50$	
1st order: $k_d = \ln(1/a)/t$	$0.223/10 = 0.0223$ h^{-1}	$0.693/40 = 0.0173$ h^{-1}	✗
2nd order: $k_d = (1/a - 1)/t$	$0.25/10 = 0.025$ h^{-1}	$1.00/40 = 0.025$ h^{-1}	✓

Answer

Second order, with $k_d = 0.025$ h^{-1}: this is typical of **sintering**. For $a = 0.25$: $1/a - 1 = 3 = k_d t \Rightarrow t = 3/0.025 =$ **120 h**.

(f) Separable deactivation kinetics

Answer

Kinetics are **separable** when the catalyst's history affects the rate only through one multiplying factor, the activity, and not through the form of the rate law:

$$-r'_A = a(t) \times (-r'_A)_{\text{fresh}} = a(t)\, k(T)\, f(C_i), \qquad a(t) = \frac{-r'_A(t)}{-r'_A(t=0)}, \quad a(0) = 1,$$

with a separate decay law $r_d = -\dfrac{da}{dt} = p(a)\, k_d(T)\, h(C_i)$. **Why it is useful:** the intrinsic kinetics (on fresh catalyst) and the decay law can be measured and modelled *independently*, and then simply multiplied in the design equation. (Non-separable: $-r'_A = f(\text{history}, \text{kinetics})$, which cannot

be split.) **Raising T:** as a falls, plants increase T so that $k(T)$ rises and $a\,k(T)$, and hence the production rate, stays constant.

(g) Choosing the reactor by decay time scale

Answer

(i) **Packed bed.** Decay is very slow, so the catalyst can stay fixed for years. A PBR gives the highest conversion per mass of catalyst, and the catalyst is replaced at rare shutdowns.

(ii) **Straight-through transport reactor (riser).** Catalyst and feed enter together and pass through in seconds, so every reacting molecule sees fresh catalyst (there is no "decay disguise"). The spent catalyst goes straight to a regenerator.

(iii) **Moving bed.** The catalyst moves slowly through the reactor, is withdrawn continuously to a regenerator (coke burn-off), and returns. The activity in the bed stays steady and the plant never stops.

Rule of thumb: the faster the decay, the more the catalyst must move (years: PBR; hours to days: moving bed; seconds: STTR or fluidised riser).

(h) Classifying deactivation

Answer

(i) **Sintering** (thermal degradation): crystallites grow, and the Ni area and the support area are lost. **Irreversible**, so it is prevented rather than cured. Remedy: run at a lower T, or use a thermally stable support ($Al_2O_3 > SiO_2 > C$), a non-oxidising atmosphere, a porous support, or trapping promoters (BaO, CaO, SrO).

(ii) **Poisoning** by sulfur: H_2S chemisorbs strongly on the Ni sites. Essentially **irreversible**. Remedy: remove the sulfur upstream (hydrodesulfurisation or a ZnO guard bed) before the reformer.

(iii) **Fouling/coking**: coke has two effects, *site coverage* and *pore blockage* (diffusion resistance). **Reversible**: burn the coke off in a regenerator (moving bed, FCC). It can also be reduced by co-feeding H_2 or steam, or by lowering T.

O_2: "poison" has an *operational* meaning. On Ag, adsorbed O is the reactant that makes ethylene oxide. On Ni, O_2 adsorbs more strongly than H_2 and ethylene, blocking the sites they need, so it acts as a poison.

Key idea

Lecture 10 is largely descriptive, so expect at least one question like (f)–(h): know the four reactor types, the three deactivation mechanisms, and which ones are reversible.

Grading guide

(a)–(e) 3 pts each, (f)–(g) 3 pts, (h) 4 pts. (a): 2 for the factor of 2, 1 for the reason. (b): 1 per correct conclusion. (c): 1 + 1 for the names, 1 for the reasoning. (d): 1 per correct match with a reason. (e): 1 for the test, 1 for k_d and the mechanism, 1 for 120 h. (f): 1 for the definition, 1 for

the rate law with $a(t)$, 1 for why it is useful and why T is raised. (g): 1 per correct reactor with a reason. (h): 1 per correctly classified case (mechanism, reversibility, remedy), 1 for the O_2 sentence.

Problem 2: Ozone decomposition (15 pts)

(a) PSSH derivation

Step 1. Rate of each step, written for O (every k is w.r.t. O, and every coefficient of O is 1):

$$r_{1,O} = +k_1 C_{O_3}, \qquad r_{2,O} = -k_2 C_{O_2} C_O, \qquad r_{3,O} = -k_3 C_O C_{O_3}.$$

Step 2. PSSH on O:

$$r_O = k_1 C_{O_3} - k_2 C_{O_2} C_O - k_3 C_O C_{O_3} = 0 \quad \Rightarrow \quad \boxed{C_O = \frac{k_1 C_{O_3}}{k_2 C_{O_2} + k_3 C_{O_3}}}$$

Step 3. Net rate of O_3, converting each step with relative rates (one O_3 per O in every step):

- step 1 consumes O_3: $r_{1,O_3} = -r_{1,O} = -k_1 C_{O_3}$;
- step 2 *forms* O_3: $r_{2,O_3} = -r_{2,O} = +k_2 C_{O_2} C_O$;
- step 3 consumes O_3: $r_{3,O_3} = r_{3,O} = -k_3 C_O C_{O_3}$.

$$-r_{O_3} = k_1 C_{O_3} - k_2 C_{O_2} C_O + k_3 C_O C_{O_3}.$$

Step 4. Simplify with the PSSH. From Step 2, $k_1 C_{O_3} - k_2 C_{O_2} C_O = k_3 C_O C_{O_3}$. Therefore

$$-r_{O_3} = 2k_3 C_O C_{O_3} = 2k_3 C_{O_3} \cdot \frac{k_1 C_{O_3}}{k_2 C_{O_2} + k_3 C_{O_3}}.$$

Answer

$$-r_{O_3} = \frac{2k_1 k_3 \, C_{O_3}^2}{k_2 C_{O_2} + k_3 C_{O_3}}$$

Check: $r_{O_2} = k_1 C_{O_3} - k_2 C_{O_2} C_O + 2k_3 C_O C_{O_3} = 3k_3 C_O C_{O_3} = \frac{3}{2}(-r_{O_3})$, which is consistent with $2\,O_3 \longrightarrow 3\,O_2$. ✓

Common mistake

(1) Forgetting that step 3 *also* consumes O_3, which gives $-r_{O_3} = k_3 C_O C_{O_3}$ and loses the factor 2. (2) Forgetting that step 2 *regenerates* O_3. Write the O_3 contribution of every step explicitly before simplifying.

(b) Excess O_2, and the start of the run

Answer

Excess O_2: $k_2 C_{O_2} \gg k_3 C_{O_3}$ (O atoms recombine with O_2 much more often than they attack O_3), so

$$-r_{O_3} \approx \frac{2k_1 k_3}{k_2}\frac{C_{O_3}^2}{C_{O_2}}, \qquad k_{obs} = \frac{2k_1 k_3}{k_2}.$$

Start of a run with pure ozone: $C_{O_2} \approx 0$, so $k_3 C_{O_3} \gg k_2 C_{O_2}$ and $-r_{O_3} \approx 2k_1 C_{O_3}$, which is **first order**. As O_2 builds up, the product **inhibits** the reaction (O_2 is in the denominator).

(c) Apparent activation energy

$k_{obs} = 2k_1k_3/k_2 = 2\dfrac{A_1A_3}{A_2}\exp\left[-\dfrac{E_1+E_3-E_2}{RT}\right]$, so

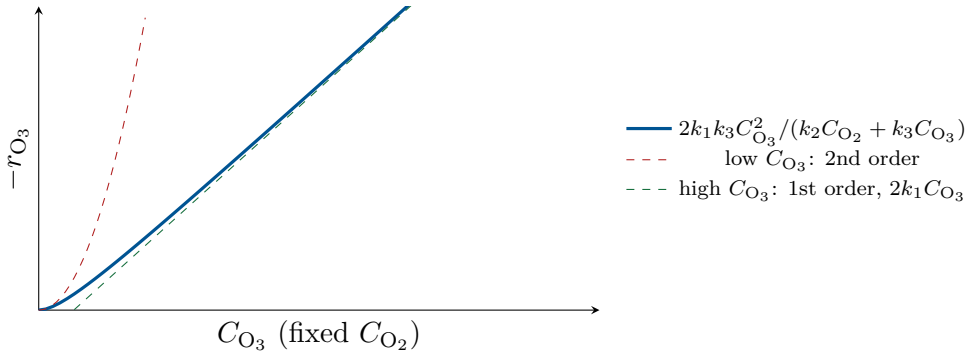
$$E_{app} = E_1 + E_3 - E_2 = 100 + 20 - 2 = 118 \text{ kJ/mol}.$$

$$\frac{k_{obs}(310)}{k_{obs}(300)} = \exp\left[\frac{118{,}000}{8.314}\left(\frac{1}{300}-\frac{1}{310}\right)\right] = \exp\left[(14{,}193)(1.0753\times10^{-4})\right] = e^{1.526}.$$

Answer

$E_{app} = 118$ kJ/mol, and k_{obs} increases by a factor of **4.60**. (If E_2 were larger than $E_1 + E_3$, then $E_{app} < 0$, as in the NO oxidation of Lec. 6.)

(d) Sketch



Answer

Second order at low C_{O_3} (with C_{O_2} fixed), changing to first order at high C_{O_3}. At high C_{O_3}, every O atom formed reacts with O_3, so the rate is set by **step 1**, the unimolecular decomposition $O_3 \longrightarrow O_2 + O$. This is the same behaviour as the Lindemann example in Lec. 6.

Grading guide

(a) 1 for the step rates written for O, 2 for the PSSH and C_O, 2 for the net O_3 rate with all three steps, 1 for the final simplification. (b) 1 for the condition, 2 for k_{obs}, 1 for the initial first order. (c) 2 for E_{app}, 1 for the factor. (d) 1 for the sketch with orders, 1 for step 1.

Problem 3: Enzyme kinetics (20 pts)

(a) Uncompetitive inhibition

Mechanism: $\mathrm{E} + \mathrm{S} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \mathrm{E\cdot S} \xrightarrow{k_3} \mathrm{E} + \mathrm{P}$, $\mathrm{E\cdot S} + \mathrm{I} \rightleftharpoons \mathrm{I\cdot E\cdot S}$ (inactive, K_I).

Step 1. PSSH on E·S. The inhibitor step is at equilibrium, so its net rate is zero: $0 = k_1 C_E C_S - (k_2 + k_3) C_{E\cdot S} \;\Rightarrow\; C_{E\cdot S} = \dfrac{C_E C_S}{K_M}$, with $K_M = \dfrac{k_2 + k_3}{k_1}$.

Step 2. Inhibitor equilibrium: $C_{I\cdot E\cdot S} = \dfrac{C_I C_{E\cdot S}}{K_I} = \dfrac{C_I C_S C_E}{K_I K_M}$.

Step 3. Total enzyme balance:

$$C_{E,t} = C_E + C_{E\cdot S} + C_{I\cdot E\cdot S} = C_E \left[1 + \frac{C_S}{K_M}\left(1 + \frac{C_I}{K_I}\right)\right].$$

Step 4. Rate, with $V_{\max} = k_3 C_{E,t}$:

$$-r_S = k_3 C_{E\cdot S} = \frac{k_3 C_S C_{E,t}/K_M}{1 + \frac{C_S}{K_M}\left(1 + \frac{C_I}{K_I}\right)} = \frac{V_{\max} C_S}{K_M + C_S\left(1 + \frac{C_I}{K_I}\right)}.$$

Step 5. Invert:

Answer

$$\frac{1}{-r_S} = \frac{1}{V_{\max}}\left(1 + \frac{C_I}{K_I}\right) + \frac{K_M}{V_{\max}} \cdot \frac{1}{C_S}$$

The slope $K_M/V_{\max}$ is **unchanged** and the intercept is multiplied by $(1 + C_I/K_I)$, so the lines are parallel.

(b) $V_{\max}$ and K_M

$1/C_S$ (L/mmol)	1.00	0.50	0.20	0.10	0.05
$1/(-r_S)$, no I (L·min/mmol)	3.00	1.75	1.00	0.75	0.625
$1/(-r_S)$, with I	4.00	2.75	2.00	1.75	1.63

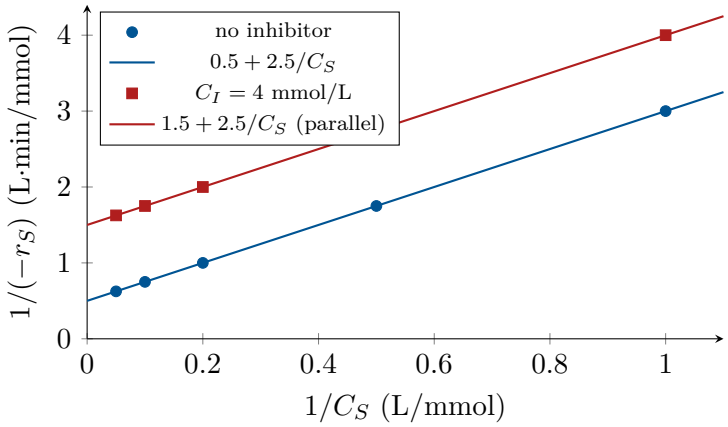
The uninhibited points lie on a straight line. Using the two end points:

$$\text{slope} = \frac{3.00 - 0.625}{1.00 - 0.05} = 2.50 \text{ min}, \qquad \text{intercept} = 0.625 - 2.50(0.05) = 0.500 \text{ L·min/mmol}.$$

(Check with a middle point: $0.5 + 2.5(0.2) = 1.00$. ✓)

Answer

$V_{\max} = 1/0.500 =$ **2.0 mmol/(L·min)**; $K_M = \text{slope} \times V_{\max} = 2.50 \times 2.0 =$ **5.0 mmol/L**.



(c) Type of inhibition and K_I

With the inhibitor: slope $= \dfrac{4.00 - 1.63}{0.95} = 2.50$ (unchanged) and intercept $= 4.00 - 2.50 = 1.50$ (tripled).

Answer

The slope is unchanged and the intercept goes up, so this is **uncompetitive** inhibition. $1 + \dfrac{C_I}{K_I} = \dfrac{1.50}{0.50} = 3 \;\Rightarrow\; K_I = \dfrac{C_I}{2} =$ **2.0 mmol/L**.

(d) Batch reactor

Mole balance at constant volume: $\dfrac{dC_S}{dt} = r_S$. Write $\phi = 1 + C_I/K_I$ ($\phi = 1$ with no inhibitor):

$$\frac{dC_S}{dt} = -\frac{V_{\max} C_S}{K_M + \phi C_S} \;\Rightarrow\; -\int_{C_{S0}}^{C_S} \left(\frac{K_M}{C_S} + \phi\right) dC_S = V_{\max} t \;\Rightarrow\; K_M \ln\frac{C_{S0}}{C_S} + \phi\,(C_{S0} - C_S) = V_{\max} t.$$

With $C_S = C_{S0}(1 - X)$:

$$t = \frac{K_M}{V_{\max}} \ln\frac{1}{1 - X} + \frac{\phi\, C_{S0} X}{V_{\max}}.$$

(i) No inhibitor, $\phi = 1$: $t = \dfrac{5.0}{2.0} \ln 10 + \dfrac{(20)(0.9)}{2.0} = 5.76 + 9.00 = 14.8$ min.

(ii) With inhibitor, $\phi = 3$: $t = 5.76 + 3(9.00) = 5.76 + 27.0 = 32.8$ min.

Answer

$t = 14.8$ min without inhibitor and $t = 32.8$ min with inhibitor. Only the zero-order (saturated) term is multiplied by ϕ. When C_S is small, most of the enzyme is free E, and an uncompetitive inhibitor can bind only E·S, so the low-C_S (first-order) rate $V_{\max} C_S/K_M$ is unaffected.

Common mistake

Using the CSTR equation, or plugging the *initial* rate into $t = C_{S0}X/(-r_S)$. A batch reactor must be integrated. For a competitive inhibitor it would be the ln term that is multiplied by ϕ instead.

Grading guide

(a) 2 for the complexes in terms of C_E, 2 for the enzyme balance, 1 for the rate law, 1 for the L–B form and its interpretation. (b) 2 for the reciprocals, 1 for the slope and intercept, 2 for $V_{\max}$ and K_M with units. (c) 2 for the type with a reason, 2 for K_I. (d) 2 for the derivation, 1 + 1 for the times, 1 for the explanation. Answers within ±3% from reading a hand-drawn line are full credit.

Problem 4: Catalytic dehydration (40 pts)

(a) Surface reaction rate-limiting

Step 1. RLS (irreversible, dual site, needs a vacant neighbour): $-r'_A = r_S = k_S C_{\mathrm{AS}} C_v$.

Step 2. Fast steps at equilibrium:

$$\frac{r_{AD}}{k_A} = P_A C_v - \frac{C_{\mathrm{AS}}}{K_A} \approx 0 \Rightarrow C_{\mathrm{AS}} = K_A P_A C_v; \qquad C_{\mathrm{BS}} = K_B P_B C_v; \qquad C_{\mathrm{WS}} = K_W P_W C_v.$$

Step 3. Site balance: $C_t = C_v + C_{\mathrm{AS}} + C_{\mathrm{BS}} + C_{\mathrm{WS}} = C_v\,(1 + K_A P_A + K_B P_B + K_W P_W)$.

Step 4. Substitute: $r_S = k_S K_A P_A C_v^2$.

Answer

$$-r'_A = \frac{k\,P_A}{(1 + K_A P_A + K_B P_B + K_W P_W)^2}, \qquad k = k_S K_A C_t^2.$$

(b) Adsorption of A rate-limiting

Step 1. RLS: $-r'_A = r_{AD} = k_A \left(P_A C_v - \dfrac{C_{\mathrm{AS}}}{K_A} \right)$.

Step 2. The surface reaction is fast and irreversible: $\dfrac{r_S}{k_S} = C_{\mathrm{AS}} C_v \approx 0 \Rightarrow C_{\mathrm{AS}} \approx 0$. Every A that adsorbs reacts immediately, so A does not accumulate on the surface. B and W are still at desorption equilibrium: $C_{\mathrm{BS}} = K_B P_B C_v$, $C_{\mathrm{WS}} = K_W P_W C_v$.

Step 3. Site balance: $C_t = C_v(1 + K_B P_B + K_W P_W)$.

Answer

$$-r'_A = \frac{k_A C_t P_A}{1 + K_B P_B + K_W P_W}.$$

Initial rates ($P_B = P_W = 0$): case (a) $-r'_{A0} = \dfrac{k P_{A0}}{(1 + K_A P_{A0})^2}$; case (b) $-r'_{A0} = k_A C_t P_{A0}$ (linear).

Common mistake

Writing $C_{\mathrm{AS}} = C_{\mathrm{BS}}/K_S$ as in the reversible Example 2 of Lec. 8. For an irreversible fast step, there is no equilibrium constant, and the fast consumption drives $C_{\mathrm{AS}} \to 0$.

(c) Model discrimination and parameters

The rate rises, passes through a maximum at $P_{A0} = 2$ atm, then falls. Only the dual-site surface-reaction law (a) has a maximum; case (b) is a straight line. **So (a) is consistent.**

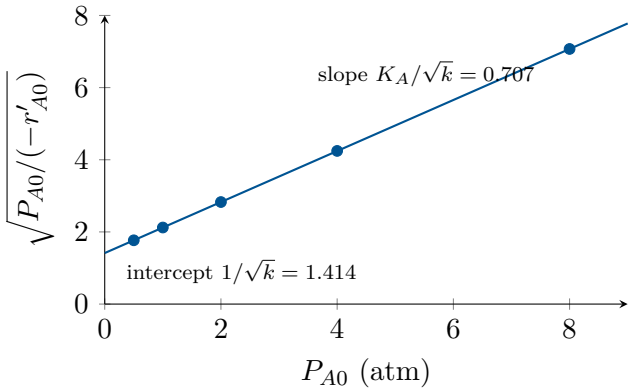
Linearise (a): $\dfrac{P_{A0}}{-r'_{A0}} = \dfrac{(1+K_A P_{A0})^2}{k} \;\Rightarrow\; \sqrt{\dfrac{P_{A0}}{-r'_{A0}}} = \dfrac{1}{\sqrt{k}} + \dfrac{K_A}{\sqrt{k}} P_{A0}.$

P_{A0} (atm)	0.5	1.0	2.0	4.0	8.0
$P_{A0}/(-r'_{A0})$	3.125	4.505	8.000	18.02	50.00
$\sqrt{P_{A0}/(-r'_{A0})}$	1.768	2.122	2.828	4.245	7.071

$$\text{slope} = \frac{7.071 - 1.768}{8.0 - 0.5} = 0.7071, \qquad \text{intercept} = 1.768 - 0.7071(0.5) = 1.414.$$

Answer

$k = \dfrac{1}{(1.414)^2} = \mathbf{0.50\ mol/(kg\ cat{\cdot}s{\cdot}atm)}$, $K_A = \text{slope} \times \sqrt{k} = 0.7071 \times 0.7071 = \mathbf{0.50\ atm^{-1}}$.
Check: the maximum should be at $1/K_A = 2$ atm with $-r'_{\max} = k/(4K_A) = 0.25$, exactly as measured. ✓



(d) Fluidised-bed CSTR

Step 1. Stoichiometry. Pure A, A $\longrightarrow$ B + W: $\delta = 1 + 1 - 1 = 1$, $\varepsilon = y_{A0}\delta = 1$. Isothermal, no ΔP:

$$P_A = P_0 \frac{1 - X}{1 + X}.$$

Step 2. With weak product adsorption, the rate is $-r'_A = \dfrac{kP_A}{(1+K_A P_A)^2}$, evaluated at the **exit** (CSTR).

(i) $P_0 = 2$ atm, $X = 0.5$: $P_A = 2(0.5)/1.5 = 0.667$ atm.

$$-r'_A = \frac{0.5(0.667)}{(1+0.333)^2} = \frac{0.3333}{1.778} = 0.1875 \text{ mol/(kg·s)}, \qquad W = \frac{F_{A0}X}{-r'_A} = \frac{(10)(0.5)}{0.1875}.$$

Answer

(i) $W = \mathbf{26.7\ kg}$.

(ii) $W = F_{A0}X/(-r'_A)_{\text{exit}}$, so the minimum W needs the maximum exit rate. From (c), the rate is largest at $P_A = 1/K_A = 2$ atm, where $-r'_A = k/(4K_A) = 0.25$ mol/(kg·s). The exit $P_A = 2$ atm at $X = 0.5$ requires

$$P_0 = P_A\frac{1+X}{1-X} = 2\cdot\frac{1.5}{0.5} = 6\text{ atm}, \qquad W_{\min} = \frac{(10)(0.5)}{0.25} = 20.0\text{ kg}.$$

Answer

(ii) $P_0 = \mathbf{6\ atm}$ and $W_{\min} = \mathbf{20.0\ kg}$. At higher pressure (e.g. $P_0 = 12$ atm, so $P_A = 4$ atm, $-r'_A = 0.222$, $W = 22.5$ kg) adsorbed A crowds the surface. The dual-site step needs a *vacant* neighbouring site, and C_v falls faster than C_{AS} rises, so the rate drops.

(e) Deactivation

(i) Second-order sintering: $a = \dfrac{1}{1+k_d t} = \dfrac{1}{1+(0.0125)(80)} = 0.50$. With the same P_0 and X, the exit P_A (and hence the fresh rate) is unchanged, and $W = \dfrac{F_{A0}X}{a(-r'_A)_{\text{fresh}}}$ is fixed. So F_{A0} must scale with a:

$$F_{A0}(80\text{ h}) = a\,F_{A0}(0) = 0.5\times 10 = 5.0\text{ mol/s}.$$

(ii) Keep F_{A0}, so the same rate is needed: $a\,k(T) = k(T_0)$:

$$\exp\left[\frac{E}{R}\left(\frac{1}{T_0}-\frac{1}{T}\right)\right] = \frac{1}{a} = 2 \;\Rightarrow\; \frac{1}{T} = \frac{1}{500} - \frac{8.314\ln 2}{80{,}000} = 0.002000 - 0.0000720 = 0.0019280\text{ K}^{-1}.$$

Answer

(i) $a = 0.50$ and $F_{A0} = \mathbf{5.0\ mol/s}$. (ii) $T = \mathbf{518.7\ K}$ (an increase of about 19 K).
(iii) Sintering is activated (k_d grows exponentially with T, $E_d \approx 30$–150 kJ/mol), so the higher T makes the catalyst decay faster, which then needs an even higher T. Also acceptable: adsorption is exothermic, so K_A falls with T; side reactions or coking speed up; or there are material and selectivity limits.

Common mistake

Using $a = e^{-k_d t}$ (first order) gives $a = 0.37$. The problem states second-order decay. Also, $k(T)$ must increase by the factor $1/a$, not by a.

Grading guide

(a) 2 for the RLS rate law with C_v, 3 for the fast-step equilibria, 2 for the site balance, 1 for the final form. (b) 2 for $C_{\text{AS}} \approx 0$ with a reason, 2 for the rate law, 2 for both initial rates. (c) 2 for the discrimination argument, 2 for the linearisation, 3 for the table and line, 2 for k and K_A. (d) 2 for ε and $P_A(X)$, 3 for (i), 3 for (ii), 1 for the physical reason. (e) 3 for (i), 3 for (ii), 2 for (iii).